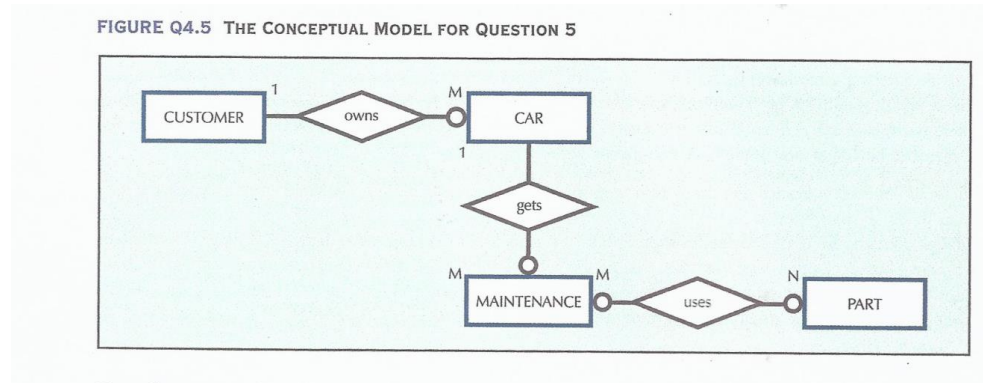


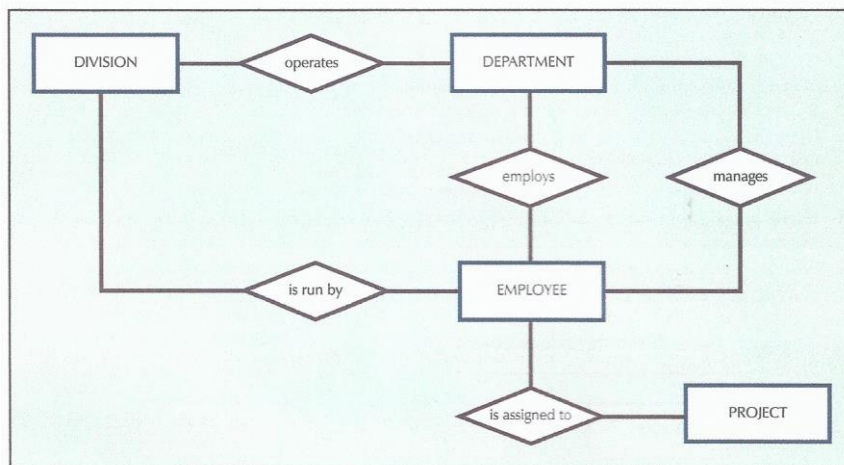
EXERCISE 3 ERD

1. Explain the difference between degree and cardinality of a relationship.



2. Given the conceptual model in figure above
 - a. Write the business rules that are reflected in it.
 - b. Identify all cardinalities
 - c. Explain how you will remove the M:N relationship between MAINTENANCE and PART

FIGURE P4.1 THE ERD FOR PROBLEMS 1–3



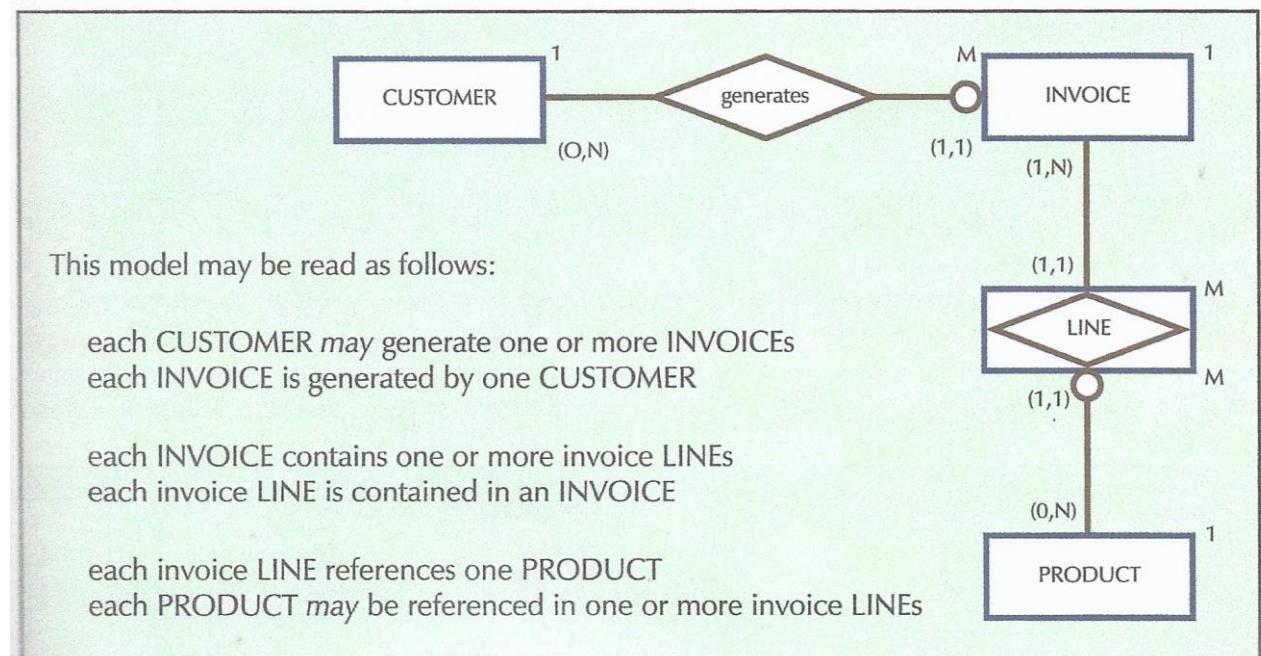
3. Use the following business rules and draw a simplified ERD
 - a. A department employs many employees, but each employee is employed by only one department
 - b. Some employees known as freelancers are not assigned to any department
 - c. An employee may be assigned to many projects and a project may have many employees assigned to it.
 - d. A project MUST have at least one employee assigned to it.
 - e. One of the employees manages each department, and each department is managed by only one employee
 - f. One of the employees runs each division, and each division is run by only one employee.

Write down all the cardinalities

Modify all m:n into 1:n relationships and rewrite connectivity and cardinalities to fit the modified relationships

Convert the Chen model into Crow's Foot model.

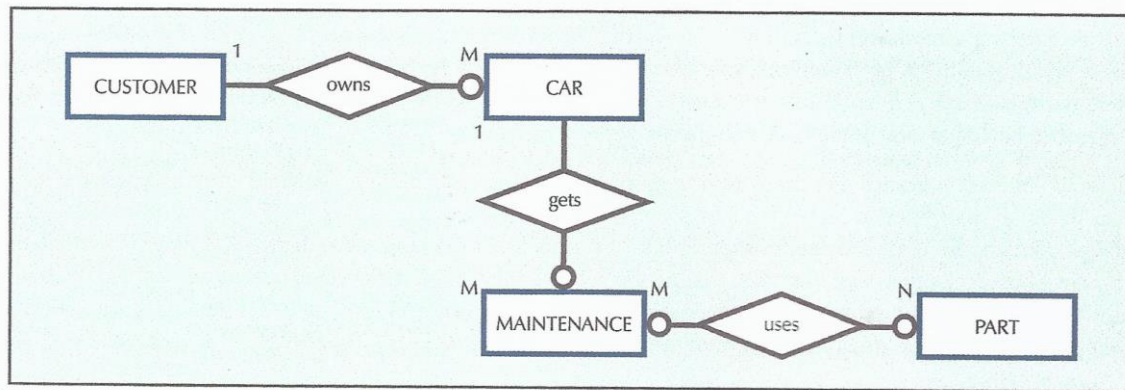
FIGURE 4.32 THE CHEN REPRESENTATION OF THE INVOICING PROBLEM



4. Create an ERD based on Crow's Foot notation
 - a. An Invoice is written by a salesrep.
 - b. Each salesrep can write many invoices, but each invoice is written for a single salesrep
 - c. The invoice is written for a single customer, however each customer may have many invoices.
 - d. The product informant is stored in a product entity
 - e. The product's vendor information I found in a vendor entity

SOLUTIONS

FIGURE Q4.5 THE CONCEPTUAL MODEL FOR QUESTION 5



This model may be read as follows:

each CUSTOMER may generate one or more INVOICES
each INVOICE is generated by one CUSTOMER

each INVOICE contains one or more invoice LINES
each invoice LINE is contained in an INVOICE

each invoice LINE references one PRODUCT
each PRODUCT may be referenced in one or more invoice LINES

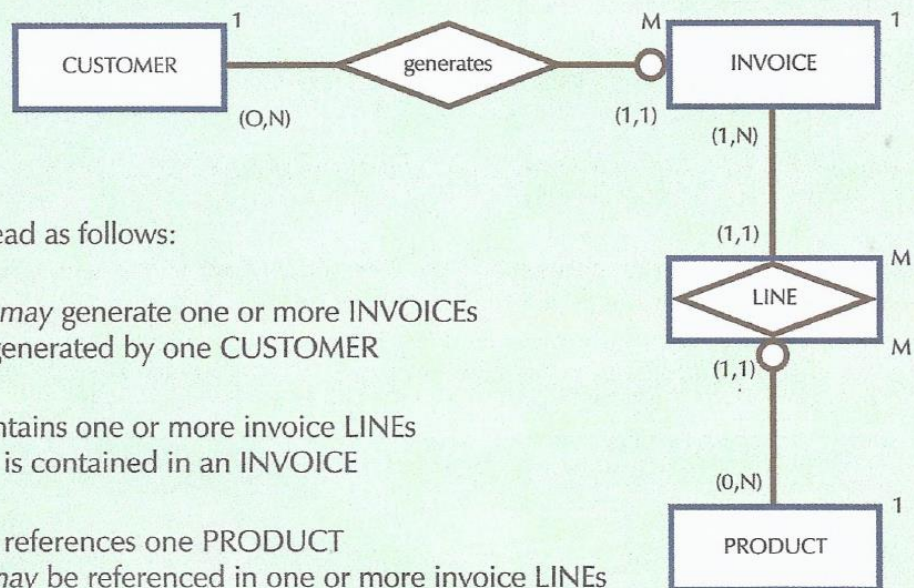


FIGURE P4.1 THE ERD FOR PROBLEMS 1–3

